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Btech cse (Sec-4)

CLASSTIME Pg. No.
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Operating System

Assignment-4

Short Answer type: Part-A

1. A race condition happens when two people try to use a shared resource at the same time, causing inconsistent results.
Example: Two people withdrawing money from the same bank account at the same time - both see ₹5000, both withdraw ₹5000, balance becomes wrong.
Mutual exclusion: Allows only one person / process to access the resource at a time, preventing conflicting actions.

- **Peterson's Solution:** Pure software; simple conceptually but works reliably only on single-core systems. Needs strict memory ordering → hardware dependent in practice.

- **Semaphores:** Supported directly by OS / hardware atomic instructions, easier to implement in real systems, works on multi-core machines.

Monitors automatically bundle data + synchronization, so locking is handled implicitly. This avoids programmer errors and reduces context-switch overhead, making concurrency safer in multi-core

4. Starvation: If writers keep arriving, readers may be blocked forever (or vice versa).
Prevention: Use fair scheduling, e.g. a queue where requests are served in order of arrival (FIFO).

5. Processes must request all resources upfront, causing:
- Low resource utilization (many resources stay idle).
- Larger waiting times → poor system performance.

Part B : Application / Numerical Based

6. Given:
S1 : $P_1 \rightarrow P_2$, $P_3 \rightarrow P_4$
S2 : $P_2 \rightarrow P_5$, $P_5 \rightarrow P_6$
S3 : $P_6 \rightarrow P_1$

(a) Global graph:
$P_1 \rightarrow P_2 \rightarrow P_5 \rightarrow P_6 \rightarrow P_1$ (cycle)
and $P_3 \rightarrow P_4$ (no cycle)

(b) Deadlock? Yes
Deadlocked processes: P_1, P_2, P_5, P_6

(c) Algorithm:
Chandy - Misra - Haas algorithm for distributed deadlock detection.

7. Local = 5 ms
Remote = 25 ms
Probability remote = 0.3
Probability local = 0.7

(a) Expected time:
$0.7 \times 5 + 0.3 \times 25$
$= 11$ ms

(b) Caching strategy:
Client-side caching with LRU → frequently accessed files stay local, reducing remote access and improving speed.

8. Full = 200ms
Incremental = 50ms
RPO = 1 second → must checkpoint atleast every 1 sec.

(a) Optimal plan (for 10 seconds):
Do 1 full checkpoint at the start + incremental checkpoint every 1 second.
Total: 1 full + 9 incremental

(b) Reasoning:
Full provides a clean base; incrementals every 1 sec keep overhead low (50 ms << 200 ms) and meet RPO.

9. (a) Challenges:
sudden traffic spikes, uneven load across regions, varying network latency.
Suitable algorithm:
Dynamic load balancing with least-loaded or Work-Stealing.

(b) Use multi-region replication + automatic failover (e.g., active-active data centres)
- RTO: very low because another region takes over instantly.
- RPO: near zero using continuous replication.